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ORIGINAL ARTICLE

Machine learning-based prediction of mortality after heart transplantation in adults with congenital heart disease: A UNOS database analysis

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Abstract

Background: Machine learning (ML) is increasingly being applied in Cardiology to predict outcomes and assist in clinical decision-making. We sought to develop and validate an ML model for the prediction of mortality after heart transplantation (HT) in adults with congenital heart disease (ACHD).

Methods: The United Network for Organ Sharing (UNOS) database was queried from 2000 to 2020 for ACHD patients who underwent isolated HT. The study cohort was randomly split into derivation (70%) and validation (30%) datasets that were used to train and test a CatBoost ML model. Feature selection was performed using SHapley Additive exPlanations (SHAP). Recipient, donor, procedural, and post-transplant characteristics were tested for their ability to predict mortality. We additionally used SHAP for explainability analysis, as well as individualized mortality risk assessment.

Results: The study cohort included 1033 recipients (median age 34 years, 61% male). At 1 year after HT, there were 205 deaths (19.9%). Out of a total of 49 variables, 10 were selected as highly predictive of 1-year mortality and were used to train the ML model. Area under the curve (AUC) and predictive accuracy for the 1-year ML model were .80 and 75.2%, respectively, and .69 and 74.2% for the 3-year model, respectively. Based on SHAP analysis, hemodialysis of the recipient post-HT had overall the strongest relative impact on 1-year mortality after HT, followed by recipient-estimated glomerular filtration rate, age and ischemic time.

Conclusions: ML models showed satisfactory predictive accuracy of mortality after HT in ACHD and allowed for individualized mortality risk assessment.

KEYWORDS
congenital heart disease, explainability, heart transplantation, machine learning, UNOS

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1 | INTRODUCTION

The life expectancy of infants born with congenital heart disease has improved dramatically in recent decades with 90% of patients now expected to reach adulthood, largely due to advances in the field of pediatric cardiology and cardiac surgery.[1] As a result, there is a growing population of adults with congenital heart disease (ACHD). Most ACHD patients remain at high risk for developing end-stage heart failure,[2] for which heart transplantation (HT) is the treatment of choice given its superior long-term outcomes and quality of life.

ACHD have inherent differences compared to other HT recipients. While they tend to be younger and have lower prevalence of comorbidities, they have higher rates of sensitization, pulmonary hypertension, and history of prior surgery, leading to an overall increased risk of short-term mortality.[3] On the other hand, risk for long-term mortality is lower in ACHD who undergo HT. These special characteristics of ACHD merit dedicated risk assessment prior to HT, and today there is a paucity of such ACHD-specific risk scores. Recently, machine learning (ML) models have been developed and validated for the prediction of 1-year mortality after HT using data from large databases with satisfactory results.[4–6] In the current study, we used data from ACHD patients undergoing HT in the United States to create and validate an ML model for the prediction of 1-year survival. We additionally performed an explainability analysis to interpret the results in a clinically relevant manner.

2 | METHODS

2.1 | Study overview

Data obtained from the United Network for Organ Sharing (UNOS) database were pre-processed to obtain the study dataset using the higher-level programming language MATLAB. The dataset was randomly split into training (70%) and validation (30%) cohorts (Figure 1). Feature selection was applied on 49 variables of the training dataset to identify the strongest predictors of mortality and improve the performance of the ML model (Table S1). Since the training cohort was not balanced regarding outcomes (i.e., more patients survived than died), a random oversampling technique was applied to guarantee equal representation. This was done by replicating patients until group sizes were equal. Selected variables were subsequently used to train CatBoost ML models for the prediction of 1- and 3-year survival.

2.2 | Data sources and inclusion-exclusion criteria

Data were obtained from the UNOS database, which collects all information that pertains to waiting list registrations and transplantations that have been listed or performed in the United States and reported to the Organ Procurement and Transplantation Network (OPTN) since October 1, 1987. The database includes baseline data, including demographics, clinical and laboratory details at the time of listing and transplantation, as well as follow-up survival data for each patient. For the present study, we conducted a retrospective analysis of the UNOS database from 2000 to 2020 to identify ACHD patients who were enlisted and underwent HT. We excluded non-adult patients, patients with history of prior transplantation, patients with simultaneous heart and other organ transplants, and patients with less than 1-year follow-up. We did not analyze variables that had >10% missing values.

2.3 | ML models and feature selection

CatBoost is an enhanced gradient-boosting ML algorithm that was used to predict mortality in the current study. Gradient boosting is a powerful ML technique that achieves state-of-the-art results in a variety of tasks, particularly those involving heterogeneous variables, noisy data and complex dependencies. CatBoost is a modification of the standard gradient boosting algorithm, which avoids prediction shifts and uses a new algorithm for categorical variables and has been shown to outperform prior implementations such as XGBoost and LightGBM in multiple applications, including medicine, meteorology, cyber security and finance.[7,8]

Feature selection was performed using Shapley Additive exPlanations (SHAP).[9] SHAP uses coalitional game theory to compute Shapley values, which express the contribution of each variable to the prediction. Instead of training a model that makes predictions using several different inputs, prediction is considered a game in which each feature (a "player") contributes to the prediction process ("score"). To determine the role of each player (feature) in determining that score (prediction), SHAP calculates the number of points that we would gain or lose in the presence or absence of the feature, respectively. The Shapley value of a specific feature is thus calculated by taking a weighted total of the differences in score between games (predictions) in which the player (feature) participates and games in which the feature is excluded. In our study, the contribution of all available variables to mortality prediction was determined via their Shapley values. We then used recursive feature elimination according to the game theory methodology described above to achieve the minimal number of features with best predictive performance. Final variables were reviewed by expert consensus (P.N.K., A.B.) before entering the CatBoost model.

2.4 | Metrics for ML model performance and explainability analysis

After training the CatBoost model with selected variables, the validation cohort was used to internally test the model and determine its predictive accuracy. We assessed the area under the receiver-operator curve (AUC), sensitivity, specificity, and positive and negative predictive value. We additionally used SHAP to assess the relative impact of CatBoost variables on 1- and 3-year mortality in order to provide

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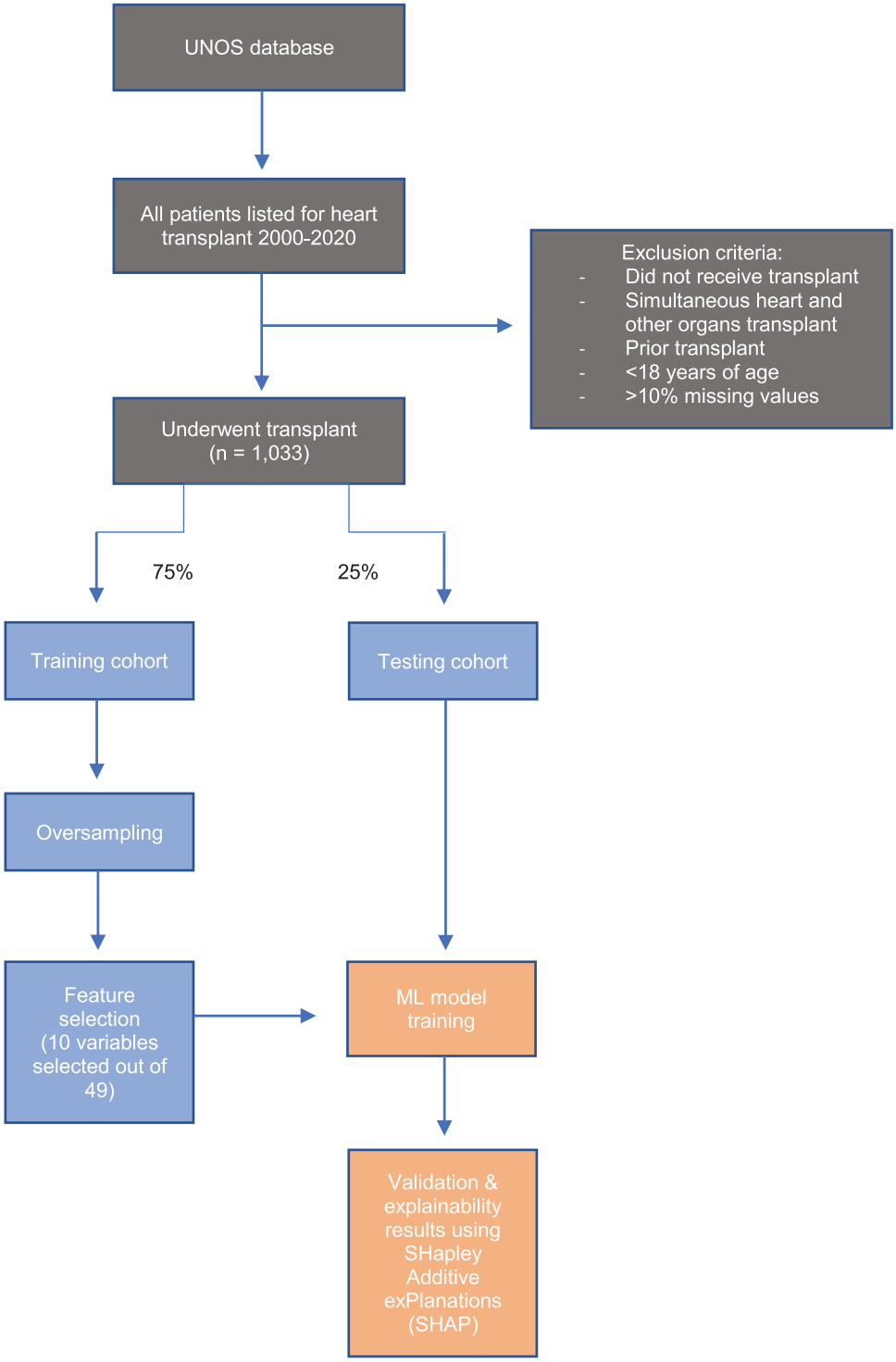

FIGURE 1 Study design for prediction of 1-year survival after heart transplantation using an ML model. Data from the UNOS database were extracted. After applying inclusion and exclusion criteria, the study dataset was randomly split in a training (70%) and a validation cohort (30%). Oversampling was performed for the HT recipients who did not survive 1-year after transplant to allow for balanced input in the ML algorithm. Features were selected using a combination of clinical expertise and a state-of-the-art feature selection algorithm, and were subsequently used to train a CatBoost decision tree model. The resulting model was tested using the validation cohort. ML, machine learning, UNOS, United Network for Organ Sharing, HT, heart transplantation

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clinical explanations of the final ML models (ML explainability analysis). Shapley values represent the contribution of each of the final ML variables to the individual prediction.

2.5 | Statistical analysis

The Shapiro-Wilk test was used to check continuous variables for normality. Variables with normal distribution are presented as mean ± standard deviation (SD) and were compared using the Student's *t*-test, whereas variables with non-normal distribution were expressed as median (interquartile range, IQR). Categorical variables were presented as frequency (%) and were compared using the Chi-square exact test. Statistical analysis was performed using MATLAB 2020b (version 9.9) and STATA MP 13.0. *p*-Values < .05 were considered as statistically significant.

3 | RESULTS

3.1 | Baseline characteristics

The UNOS-derived cohort of eligible patients consisted of 1033 recipients (median age 34, IQR 23–45 years; 61.1% male) who underwent isolated HT between 2000 and 2020. There were no significant differences between patients assigned to the development and validation cohorts (Table S2). At 1 year after HT, there were 205 (20%) deaths. Recipients who died within the first year were older (median age 37 [IQR 27–46] vs. 33 [23–45] years), slightly heavier (body mass index [BMI] 24.9 [IQR 21.1–29.1] vs. 23.6 [20.6–28.2] kg/m^2) and were more likely to require dialysis (7.32% vs. 2.66%), extra-corporeal mechanical oxygenation (ECMO; 3.90% vs. 1.33%) or mechanical ventilation (7.32% vs. 1.93%) prior to transplant. Additionally, ACHD patients that died within the 1st year after HT also had lower estimated glomerular filtration rate (eGFR) (75.5 [47.2–99.4] vs. 89.1 [61.5–113] ml/min/1.73 m^2), slightly higher total bilirubin (.90 [.60–1.18] vs. .80 [.50–1.30] mg/dL) and were more likely to have positive human immunodeficiency virus (HIV; 1.51% vs. .12%) serostatus. In terms of procedural characteristics, ischemia time was increased in those who did not survive 1-year post-HT (3.78 [3.04–4.56] vs. 3.38 [2.62–4.10] h). Following HT, ACHD patients who did not survive 1 year were more likely to have experienced stroke (9.76% vs. 1.81%, $p < .001$), as well as to require dialysis (51.7% vs. 13.2%, $p < .001$).

3.2 | Feature selection

SHAP was first used to identify the strongest predictors of mortality among the 49 available variables. It was determined via recursive feature elimination that the presence of more than 12 variables provided little additional predictive capability to the SHAP models, whereas the use of less than 10 variables significantly decreased the models' predictive accuracy. To keep the final ML model as simple and efficient as possible, a 10-variable ML model was chosen. Out of the 10 variables, seven (eGFR, ischemic time, post-operative hemodialysis, days in UNOS 1A status, post-operative stroke, positive HIV and positive HCV) were taken directly from recursive feature elimination. Three variables were selected by expert consensus (Dr. Kampaktsis, Dr. Briasoulis) namely age, total bilirubin and functional status based on their established predictive value in HT patients from prior studies.

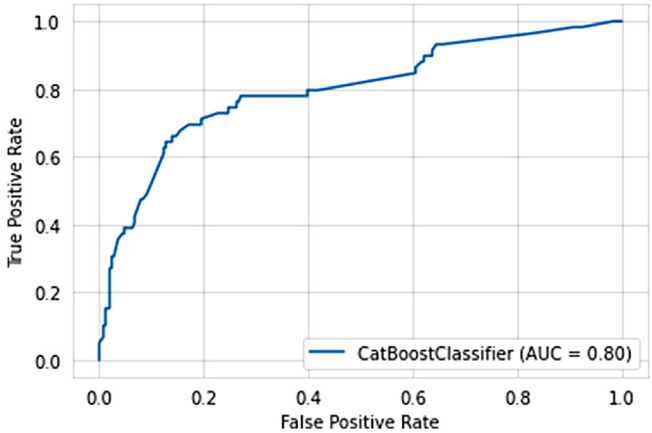

FIGURE 2 ML model performance. (A) Confusion matrix showing the predictive performance of the CatBoost decision tree ML model using 10 selected variables for 1-year mortality after HT. (B) ROC curve of the model with an AUC of .80. AUC, area under the curve, HT, heart transplantation, ML, machine learning, ROC, receiver-operating characteristic

3.3 | ML model performance and explainability

The final CatBoost ML model for 1-year mortality achieved an AUC of .80 (95% confidence interval CI: [.687, .811]), a predictive accuracy of 75.2%, a sensitivity of 75%, a specificity of 75%, a PPV of 42%, and a NPV of 93% (Figure 2). The results of explainability analysis for the ML model using SHAP are shown in Figures 3 and 4. The SHAP summary plot in Figure 3 illustrates the relative impact of each one of the 10 model variables on the outcome, with each dot representing an individual patient. Overall, post-operative hemodialysis of the recipient had the strongest relative impact on 1-year mortality after HT, followed by eGFR, age of the recipient, and ischemic time. The other top variables that were used by the ML model were total bilirubin, days in 1A status, functional status, post-operative stroke, and HIV and HCV-positive status of the recipient. The final CatBoost ML model for 3-year mortality achieved an AUC of .69, a predictive accuracy of 74.2%, a sensitivity of 51%, a specificity of 85%, a PPV of 63%, and a NPV of 78% (Figure 5).

3.4 | Examples of mortality prediction using the proposed ML model

Figure 4 illustrates two examples of predictions by the CatBoost ML model for two different patients using SHAP force plots. Patient A

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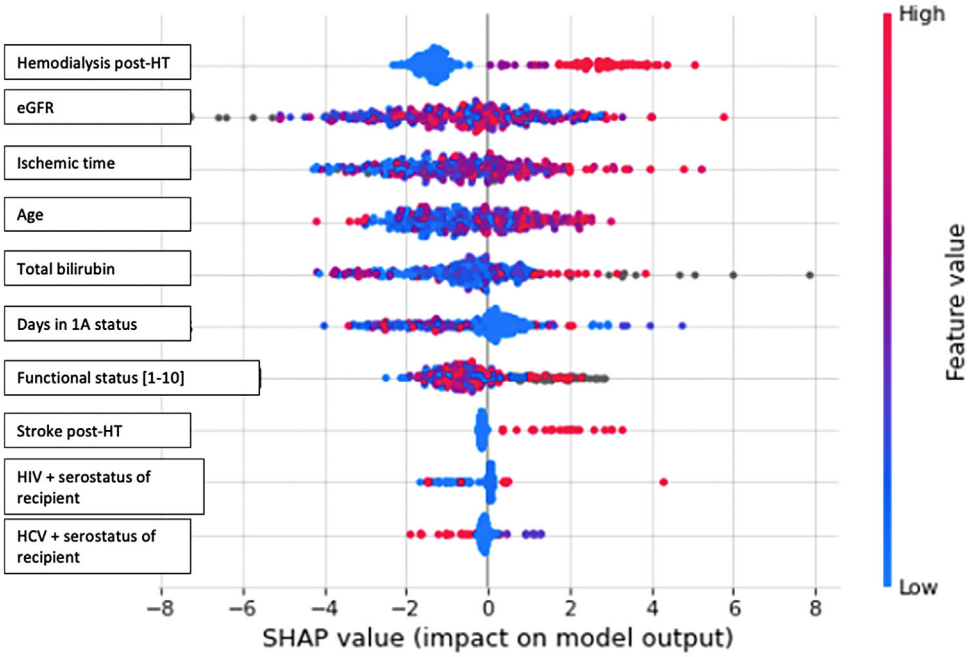

FIGURE 3 SHAP summary plot for 1-year mortality. The plot illustrates the features' impact on the output of the ML model (i.e., the predicted risk of mortality), with each dot representing an individual patient/sample. Features are sorted by the sum of SHAP value magnitudes over all samples. The color represents the feature value (blue – low; red – high), while the magnitude and direction of the effect on predicted mortality are reflected by the SHAP value. This reveals, for example, that a higher ischemic time increases the predicted risk of mortality. Similarly, the features hemodialysis post-HT, eGFR, age, total bilirubin, days in 1A status, functional status, stroke post-HT, HIV and HCV seropositive status have a positive effect on the prediction outcome (their increase drives the output to increase). SHAP, Shapley Additive exPlanations

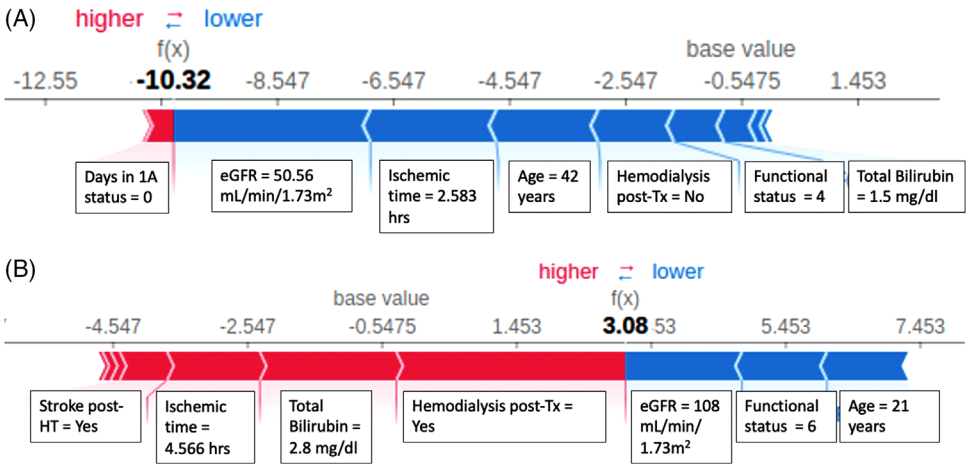

FIGURE 4 SHAP force plots. SHAP force plots allow the physician to see how features contributed to the model's prediction for a specific observation and how the model arrived at the prediction for a specific observation. The binary target was 0 for survival and 1 for death. The number in bold is the model's overall score for each patient. Higher scores lead the model to predict 1 or mortality and lower scores lead the model to predict 0 or survival. The features that were important to making the prediction for each observation are shown in red and blue, with red representing features that pushed the model score higher, and blue representing features that pushed the score lower. Features that had a bigger impact on the score are located closer to the dividing boundary between red and blue, and the size of that impact is illustrated by the size of the bar. (A) This patient was alive at 1-year post-transplant. In this case, preserved renal function (eGFR), relatively low ischemic time, age, absence of hemodialysis post-transplant (index hospitalization), good functional status and low total bilirubin were the contributing factors with this order of importance that led to a low overall score (-10.32) and the correct prediction of being alive at 1-year post-transplant. (B) This patient died within 1-year post-transplant. In this case, hemodialysis post-transplant (index hospitalization), increased total bilirubin, high ischemic time and a stroke post-transplant (index hospitalization) were the contributing factors with this order of importance that led to a high overall score (3.08) and the correct prediction of being dead at 1-year post-transplant. SHAP, Shapley Additive exPlanations

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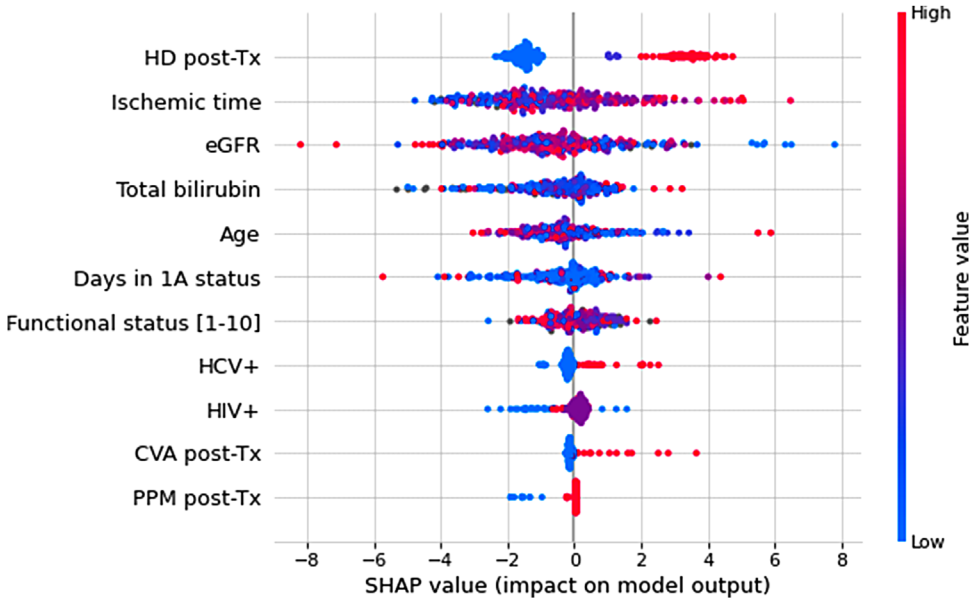

FIGURE 5 SHAP summary plot for 3-year mortality. The SHAP summary plot illustrates the features' impact on the output of the ML model (i.e., the predicted risk of 3-year mortality), with each dot representing an individual patient/sample. Features are sorted by the sum of SHAP value magnitudes over all samples. The color represents the feature value (blue – low; red – high), while the magnitude and direction of the effect on predicted mortality are reflected by the SHAP value

was alive at 1 year post-transplant. In this case, preserved renal function (eGFR), relatively low ischemic time, age, absence of hemodialysis post-transplant (index hospitalization), good functional status and low total bilirubin were the contributing factors with this order of importance that led to a low overall score (-10.32) and the correct prediction of being alive at 1-year post-transplant. Patient B died within 1-year post-transplant. In this case, hemodialysis post-transplant (index hospitalization), increased total bilirubin, high ischemic time and a stroke post-transplant (index hospitalization) were the contributing factors with this order of importance that led to a high overall score (3.08) and the correct prediction of being dead at 1-year post-transplant.

4 | DISCUSSION

The salient findings of this UNOS database analysis can be summarized as follows: (i) ML was utilized to create a 10-variable model for the prediction of 1- and 3-year mortality in ACHD patients undergoing HT with satisfactory accuracy, (ii) SHAP explainability found that eGFR, age, bilirubin, ischemic time and post-HT hemodialysis were the overall top five predictors of the 10-variable ML-derived risk score, which is in accordance with prior studies, and (iii) SHAP analysis allows for an individualized prognostication via force plots, which illustrate the relevant impact of each variable on the outcome for each individual patient.

About 3000 HTs are performed annually in the United States with 1-year survival close to 90%.[10] Several risk scores have been proposed for risk assessment.[11–13] ACHD who undergo HT represent a special group given their different clinical characteristics. In terms of outcomes, 1-year mortality risk after HT is higher in ACHD (16%–19%) compared to other patients (6%–9%), with long-term survival being more favorable.[14–16] Today, there is still a paucity of ACHD-specific mortality risk scores. Recently, Seese et al.[17] created a 13-point risk score for 1-year mortality in ACHD patients undergoing HT using the UNOS database. Results showed a significant association of the score with 1-year mortality. Previously, Burchill et al.[18] introduced a risk score based on 437 ACHD HT patients from the International Society for Heart and Lung Transplantation (ISHLT) registry, which has not been validated to our knowledge. Artificial intelligence and ML algorithms offer advantages in regards to their ability to analyze large and multivariable datasets.[19] To our knowledge, this is the first study utilizing ML methods to predict mortality after HT in ACHD. The final ML model showed reasonably good predictive accuracy with an AUC of .80.

The overall study design, involving data pre-processing, splitting into training and validation cohorts, feature selection, creation of ML models and explainability, follows a standard ML methodology that has been used by our group in prior studies.[4,20] In this study, we used the CatBoost for 1-year and 3-year mortality prediction, as this ML algorithm minimizes errors introduced by categorical variables and has been previously used multiple fields, including cardiology.[21] The CatBoost algorithm has also been specifically used in the field of organ transplantation to predict bleeding after liver transplantation.[22] In this study, we used SHAP for explainability analysis, a technique that has been used to explain ML prediction models in the pediatric transplant population.[23] We additionally used SHAP for feature selection.

SHAP explainability analysis of the ML model allowed us to assess the individual impact of the 10 variables of the CatBoost model on mortality and assess the model's clinical meaningfulness. Recipient pre-transplant eGFR and post-operative hemodialysis were strongly associated with 1-year mortality, which is in accordance to established knowledge. Kidney dysfunction is common in ACHD patients,

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particularly those with heart failure who are listed for HT and carries an incremental risk.[24] A previous analysis of the UNOS database showed an independent association of eGFR with mortality, with a 1.5-fold increased risk in those with <30 ml/min/1.73 m^2 compared with those with ≥90 ml/min/1.73 m^2. This is likely in the setting of poor baseline kidney function, exposure to surgical risk and nephrotoxic medications such as calcineurin inhibitors. Furthermore, it is clinically intuitive that post-operative hemodialysis has been associated with a two-fold increased risk of mortality.[25]

Age and total serum bilirubin have been previously shown to have a strong prognostic role in ACHD undergoing HT and were therefore selected by expert consensus in our study. Burchill et al. found that recipient age was associated with increased 1-year mortality in ACHD who underwent HT, with ages 25–30 carrying the highest risk.[3] Chronic elevations in systemic venous pressures put patients with ACHD at high risk for congestive hepatopathy, portal hypertension, and cardiac cirrhosis. Indeed, previous studies in in patients with ACHD undergoing HT have shown that the MELD-XI (Model for End Stage Liver Disease eXcluding INR) risk score predicted a composite endpoint of 90-day mortality, graft failure, or retransplantation, as well as 5-year mortality. Pre-transplant functional status has been previously associated with worse outcomes after HT in adult and pediatric patients[26] and was used by our ML model. In addition, while waiting list mortality has decreased in the United States following broader regional sharing of donors, the risk of death while waiting is not trivial and increases with days spent in UNOS 1A status.[27] Post-operative stroke is a rare complication after HT, however it has been recently identified as a predictor of long-term outcomes in the HT population.[28] Finally, HIV and HCV represent chronic viral diseases for which effective medical treatments now exist. However, mortality remains increased in these patients compared to the general population.[29]

In the current study, we did not perform a head-to-head comparison of the ML-derived prediction models with other risk scores that were derived using traditional statistics. Therefore, whether the current ML-derived models outperform previously derived risk scores for prediction of mortality after HT in patients with ACHD remains to be seen. However, we do note a significant advantage of the specific ML-based methodology with clinical implications: SHAP explainability analysis allows for individualization of prognostic evaluation by force plots, as shown in Figure 4. For every patient with ACHD who undergoes HT, SHAP force plots break down the impact of each variable that contributes to the overall prediction. In contrast to traditional statistics where each variable has a set "weight" on the final prediction equation, the current ML approach allows for a greater insight to the contribution of each variable for different patients. For patient A in Figure 4, renal function and ischemic time were the two most important variables that pushed the prediction towards survival, whereas for patient B renal function still had a protective role but ischemic time was increased and therefore detrimental. Specifically, we see that for patient B, good renal function and functional status as well as young age were not enough to predict a good outcome. Post-HD dialysis, high total bilirubin and ischemic time as well as a post-HT stroke weighed heavily towards correctly predicting a poor outcome. This individualized approach provides clinically meaningful insight to the treating physician.

4.1 | Study limitations

Although the UNOS database is currently used broadly for epidemiologic research and serves many aims, it does not provide the exact type of CHD or the anatomical and physiological classification of each patient. We also recognize that high predictive accuracy has been difficult to achieve using UNOS and concerns about the database have been raised.[30] Furthermore, we used post-operative complications of the index hospitalization, namely dialysis and stroke in this analysis, which are not available at baseline. Additionally, our results will require external validation. Finally, although one of the main disadvantages of ML is the lack of transparency and explainability, this was addressed in the current study with the use of SHAP analysis.

5 | CONCLUSIONS

In conclusion, this study developed, for the first time, ML-derived risk models for the prediction of mortality in ACHD patients undergoing HT. Results were satisfactory. The relative impact of the 10 variables used by the ML model was examined by explainability analysis and was evaluated as clinically meaningful. Improvement in prognostication and understanding of the individual predictors of mortality may enable the implementation of better strategies to optimally select recipients and donors, and to eventually improve outcomes for the growing population of ACHD patients who undergo HT. External validation of the models in other HT cohorts and assessment of its predictive capacity of long-term survival after HT are warranted.

CONFLICTS OF INTEREST

None.

DATA AVAILABILITY STATEMENT

Data are available upon written request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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